Code No: 138GY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, December - 2020
SENSORS AND TRANSDUCERS
(Common to EEE, ECE, CSE, IT)

Time: 2 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. Explain the three main functional elements in a measurement system in detail and give their significances. [15]

2.a) Explain systematic error in detail. How can it be minimized?
b) List and explain the factors to be considered for selecting a transducer. [7+8]

3. List and explain the dynamic characteristics of an Instrument. [15]

4. Explain in detail about the dynamic response of Zero order instruments. [15]

5. Explain with the help of a neat diagram the working of a load cell. State its applications. [15]

6. State the main advantages and disadvantages of semiconductor strain gauge compared to the metallic wire strain gauge. [15]

7. Explain the working and construction of a rotational variable differential transducer (RVDT). [15]

8.a) List the properties of piezo – electric crystals
b) Explain how the measurement of displacement can be done using Hall effect transducer. [6+9]

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JNTUH Used Papers 2020